

Commerce XI Maths App

Theory (Pages 16–18)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 25

Question 1 - Theory P16

An ogive is a

Options

- A. cumulative frequency curve
- B. frequency polygon
- C. bar diagram
- D. pie chart

Correct answer: cumulative frequency curve

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives cumulative frequency curve.

Question 2 - Theory P16

The two types of ogives are

Options

- A. less than ogive and more than ogive
- B. upper and lower ogive
- C. positive and negative ogive
- D. simple and compound ogive

Correct answer: less than ogive and more than ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives less than ogive and more than ogive.

Question 3 - Theory P16

For a less than ogive, plot upper class boundaries against

Options

- A. less than cumulative frequencies
- B. class frequencies
- C. more than cumulative frequencies
- D. class marks

Correct answer: less than cumulative frequencies

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives less than cumulative frequencies.

Question 4 - Theory P16

For a more than ogive, plot lower class boundaries against

Options

- A. more than cumulative frequencies
- B. less than cumulative frequencies
- C. class marks
- D. relative frequencies

Correct answer: more than cumulative frequencies

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives more than cumulative frequencies.

Question 5 - Theory P16

The curve obtained by joining cumulative-frequency points smoothly is called an

Options

- A. ogive
- B. histogram
- C. frequency polygon
- D. sector diagram

Correct answer: ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives ogive.

Question 6 - Theory P17

To locate the median on an ogive, draw a horizontal line from

Options

- A. $N/2$ on the Y-axis
- B. $N/4$ on the Y-axis
- C. $2N$ on the X-axis
- D. N on the X-axis

Correct answer: $N/2$ on the Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $N/2$ on the Y-axis.

Question 7 - Theory P17

To locate Q_i on an ogive, draw a horizontal line from

Options

- A. $(i N)/(4)$ on the Y-axis
- B. $(i N)/(10)$ on the Y-axis
- C. $(i N)/(100)$ on the Y-axis
- D. $(N)/i$ on the X-axis

Correct answer: $(i N)/(4)$ on the Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(4)$ on the Y-axis.

Question 8 - Theory P17

To locate D_i on an ogive, draw a horizontal line from

Options

- A. $(i N)/(10)$ on the Y-axis
- B. $(i N)/(4)$ on the Y-axis
- C. $(i N)/(100)$ on the Y-axis
- D. $10N/i$ on the X-axis

Correct answer: $(i N)/(10)$ on the Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(10)$ on the Y-axis.

Question 9 - Theory P17

To locate P_i on an ogive, draw a horizontal line from

Options

- A. $(i N)/(100)$ on the Y-axis
- B. $(i N)/(10)$ on the Y-axis
- C. $(i N)/(4)$ on the Y-axis
- D. $100N/i$ on the X-axis

Correct answer: $(i N)/(100)$ on the Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(100)$ on the Y-axis.

Question 10 - Theory P17

The intersection point of less than and more than ogives locates the

Options

- A. median
- B. mode
- C. mean
- D. range

Correct answer: median

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives median.

Question 11 - Theory P17

Partition values obtained graphically are generally

Options

- A. approximate
- B. always exact
- C. always integers
- D. undefined

Correct answer: approximate

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives approximate.

Question 12 - Solved Example P17

From the textbook marks ogive, Q_2 is approximately

Options

- A. 27.5
- B. 25
- C. 30
- D. 40

Correct answer: 27.5

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 27.5.

Question 13 - Solved Example P17

From the textbook marks ogive, D_8 is approximately

Options

- A. 40
- B. 27.5
- C. 30
- D. 50

Correct answer: 40

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 40.

Question 14 - Solved Example P17

From the textbook marks ogive, P_{60} is approximately

Options

- A. 30
- B. 27.5
- C. 40
- D. 60

Correct answer: 30

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 30.

Question 15 - Solved Example P18

Before plotting discontinuous classes 10–14, 15–19, ..., they are converted to

Options

- A. continuous classes 9.5–14.5, 14.5–19.5, ...
- B. class marks only
- C. percentages
- D. descending frequencies

Correct answer: continuous classes 9.5–14.5, 14.5–19.5, ...

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives continuous classes 9.5–14.5, 14.5–19.5,

Question 16 - Solved Example P18

From the overtime-work ogive, Q_1 is approximately

Options

- A. 18
- B. 11
- C. 23
- D. 25

Correct answer: 18

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 18.

Question 17 - Solved Example P18

From the overtime-work ogive, the median is approximately

Options

- A. 23
- B. 18
- C. 25
- D. 29

Correct answer: 23

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 23.

Question 18 - Solved Example P18

From the overtime-work ogive, P_{70} is approximately

Options

- A. 25
- B. 18
- C. 23
- D. 30

Correct answer: 25

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 25.

Question 19 - Solved Example P18

From the overtime-work ogive, employees working less than 11 hours are approximately

Options

- A. 4
- B. 6
- C. 11
- D. 20

Correct answer: 4

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 4.

Question 20 - Theory P17

After meeting the ogive, a perpendicular is drawn to the

Options

- A. X-axis to read the partition value
- B. Y-axis to read total frequency
- C. origin only
- D. class-frequency column

Correct answer: X-axis to read the partition value

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives X-axis to read the partition value.

Question 21 - Theory P16

A zero cumulative frequency is assigned before the first class when drawing a

Options

- A. less than ogive
- B. more than ogive
- C. histogram
- D. pie chart

Correct answer: less than ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives less than ogive.

Question 22 - Theory P16

A zero cumulative frequency is assigned after the last class when drawing a

Options

- A. more than ogive
- B. less than ogive
- C. frequency polygon
- D. bar graph

Correct answer: more than ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives more than ogive.

Question 23 - Theory P17

The horizontal level used to locate Q_1 is

Options

- A. $N/4$
- B. $N/2$
- C. $(3N)/(4)$
- D. $N/10$

Correct answer: $N/4$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $N/4$.

Question 24 - Theory P17

The horizontal level used to locate Q_3 is

Options

- A. $(3N)/(4)$
- B. $N/4$
- C. $N/2$
- D. $(3N)/(10)$

Correct answer: $(3N)/(4)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(3N)/(4)$.

Question 25 - Let's Think P18

The overtime example can also be solved by drawing a

Options

- A. more than ogive
- B. pie chart only
- C. simple bar chart
- D. scatter plot

Correct answer: more than ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives more than ogive.